

Time is geometry: an operator foundation model that learns the shape of dynamical systems

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Time-Series Operator project — preprint

Abstract

Foundation models for time series currently adapt language architectures: signals are chopped into tokens and modelled as sentences. Time is not text. A time series is a sequence of discrete observations of a continuous, high-dimensional and often chaotic reality. We present the Time-Series Operator (TSO), a geometry-first architecture that never tokenizes: it reconstructs phase space from a single observed channel (Takens’ embedding), linearizes the flow in Koopman coordinates (both a closed-form Fourier lift and a learned autoencoder lift), reads the attractor’s health through a data-driven Lyapunov tipping detector, and – as a foundation model – pretrains one shared operator across 23 series from 8 physical and economic domains using four self-supervised temporal pretexts: phase-space reconstruction, linear latent dynamics, scale covariance (renormalization-group consistency under $2\times$ decimation) and the arrow of time (a convolutional head classifies forward vs. reversed latent trajectories at 94–97% accuracy). A frozen zero-shot probe (closed-form linear fit in the pretrained latent, zero gradient steps) forecasts held-out systems and beats a per-series GRU autoregressive token baseline across every scale at which we compare: a median +68 percentage-point skill advantage on the local 23-series study (16/23 wins, two-sided binomial $p=0.09$) and +46.5 points in the scaled in-kernel comparison over 40 series (28/40 wins, $p=0.017$; a GPU width-doubled variant reaches 29/40, and the final architecture — a regime-balanced corpus with the Koopman pretext re-weighted — reaches 31/40 wins with the first *positive* median skill, +1.8 points over persistence). Scaling the corpus and model width from 7 to 40 series and 4k to 25k iterations raised the fraction of series with positive frozen skill from 2/7 to 13/23 (20/40 across the full 40-series evaluation); extending pretraining to 60k iterations saturates the pretext losses without further transfer gains (16/40), and making the probe a pretraining term (joint multi-step roll-out consistency plus unit-circle spectral regularization, v12) does not move the plateau (11/23 on the shared subset; v9 beats it head-to-head 16/23, $p = 0.05$). The plateau *is* broken by combining the two levers — the v13 dynamics battery re-balanced toward smooth/seasonal/spiky regimes *and* the Koopman pretext forced to weight 2.5 at v11 capacity (v14): 14/23 on the shared subset (best of any run), 31/40 wins in-kernel, 24/40 positive, median +1.8 points, and the first run to win a paired head-to-head against v9 (13/23, $p = 0.34$). Five-seed replication confirms the break is not a lucky seed: all five v14-recipe runs land at 12–14/23 on the shared subset (v9: 12/23; v11 seeds: 12/5/8) with a positive median skill in every seed (+2.0, +6.4, +0.5, +1.4, +3.8 points vs a negative v9 median), and the Wilcoxon signed-rank test on per-series medians against v9 is significant at $p = 0.03$ (pooled positive fraction 65/115 vs v9’s rate, $p = 0.20$). The direction is consistent across every seed, and the magnitude gains are statistically significant at a conventional level. A frozen Chronos-t5-small (~80k-series pretraining corpus) still beats the operator on the identical protocol (median +24.7 vs +1.8 percentage points; 24/40 head-to-head), quantifying how far corpus breadth moves the curve. Ablations show the pretexts converge but only pay off with scale. The operator’s scale-space spectrum distinguishes deterministic physics from stochastic physiology without any labels, and its tipping detector flags the Lorenz chaos onset before the fully chaotic attractor exists. Finally, a pure spectral consequence of pretraining: the frozen operator, applied to the held-out sunspot record through the renormalization (coarsening) leg,

rediscovers the ~ 11 -year solar cycle from its fitted Koopman eigenvalues, converging to 130 months against the known 132-month Schwabe period.

1 Introduction

Language models scale because text is a discrete code: tokenize, predict the next token, repeat. Time series are not a code. A sample of a chaotic or stochastic process carries no symbol identity; “next-token prediction” on numbers degenerates into copying the last value when the dynamics are unpredictable, and into memorizing local noise when they are not [1, 2]. The alternative, classical in dynamical systems and physics, is to stop predicting values entirely and instead learn the *geometry of the flow*: the manifold on which the system lives, the vector field that moves it, and the operators that linearize it.

The Time-Series Operator (TSO) implements this programme end to end, from one raw scalar channel to a pretrained, cross-domain operator:

1. **Phase space.** A single channel $x(t)$ is delay-embedded into $\mathbf{s}(t) = [x(t), x(t - \tau), \dots, x(t - (m - 1)\tau)]$ with data-driven τ (first zero of the autocorrelation / first local minimum) and m (false-nearest-neighbours collapse). Takens’ theorem guarantees a manifold topologically equivalent to the true attractor [3].
2. **Linearization.** On that manifold we fit Koopman operators [4, 5]: the exact DMD matrix [6], an extended DMD with a random-Fourier lift (258 dimensions), and a *learned* lift — an autoencoder ϕ, K, ψ whose latent makes the flow linear, trained with reconstruction, one-step dynamics, latent linearity and multi-step consistency losses.
3. **Health.** A Wolf-style largest-Lyapunov estimator [7] on the reconstructed attractor turns the model into a tipping-point detector: positive λ_1 = exponential divergence = bifurcation precursor.
4. **Scale.** The same Koopman fit at $1\times$, $2\times$, $4\times$ and $8\times$ decimation provides the spectrum across temporal lenses — the renormalization-group view of how micro-fluctuations become macro-structure.
5. **Foundation.** A shared operator Φ (encoder, one linear Koopman matrix shared across *all* domains, decoder, a scale-transfer head and an arrow-of-time head) is pretrained across domains with four self-supervised pretext tasks. At test time the lift is *frozen*: only a linear Koopman matrix is fit by least squares in the latent (rank-reduced, spectrally clipped) and the forecast is computed in closed form — zero gradient steps on the target system.

2 Results

2.1 The scale-space diagnostic separates physics from physiology

Table 1 reports the core pipeline on the Lorenz-63 butterfly (one noisy channel of a 3-D chaotic system) and on real heart-rate variability (MIT-BIH record 215, 2,500 beats, fetched through the Kaggle legacy API). The decisive row is the last one: the top Koopman eigenvalue magnitude under decimation. Deterministic convection keeps its spectrum pinned to the unit circle from $1\times$ to $8\times$ — scale-covariant physics — while a noisy heart sheds its spectral content as the lens coarsens. The *same operator*, fitted with no labels, tells a deterministic system from a stochastic one by the shape of this decay.

Table 1: Core TSO pipeline. Skill = % RMSE reduction over the persistence baseline on a held-out horizon. λ_1 = largest Lyapunov exponent.

metric	Lorenz-63 (chaos)	MIT-BIH HRV (heartbeats)
Takens delay / dimension	$\tau=59, m=3$	$\tau=1, m=6$
eDMD-RFF skill	+21.3%	+3.3%
plain DMD skill	+18.2%	+3.3%
neural vector field, pre-divergence correlation	0.79 (8 steps)	0.74
neural field loss (learned ds/dt)	0.022	0.152
top $ \lambda $ at scales $1\times \rightarrow 8\times$	0.999 $\rightarrow$ 0.937	0.69 $\rightarrow$ 0.21
Lyapunov reading	$\lambda_1 = +0.024$ (chaotic)	$\lambda_1 = +0.017$

2.2 The operator detects tipping points before they arrive

For the Lorenz system with parameter ρ swept from 1 to 40 (one observed channel, no labels), the Wolf estimator returns durably negative λ_1 in the laminar regime, then a durable crossing to $\lambda_1 > 0$ at $\rho \approx 21$ — in the transient-chaos corridor that physically precedes the homoclinic explosion at $\rho \approx 24.74$ [8]. The detector therefore functions as an *early* warning: the raw series looks normal while the geometry of the reconstructed attractor is already warping (Fig. ??).

2.3 Learned vs. fixed Koopman lifts

On a single system the fixed 258-dimensional Fourier lift is the more linear coordinate system ($1.9\times$ lower one-step latent error on held-out pairs than the learned 64-dimensional lift) and forecasts within 2 percentage points of the learned one. The learned lift’s value is not per-system accuracy but *sharing*: one ϕ that maps every domain into one latent. That is what the foundation sections below exploit.

2.4 Foundation pretraining

One FoundationOperator was pretrained on 21 training series across 8 domains and evaluated zero-shot on 2 held-out systems (sunspots, US COVID-19 daily new cases) plus all training series (Table 3). Four pretexts are optimized jointly; all converge at every scale tried (reconstruction loss 0.01–0.04, linear-dynamics 0.3–0.6, scale covariance 0.02–0.04). The arrow-of-time head — a temporal convolution over latent windows distinguishing forward from reversed trajectories — reaches 94–97% accuracy across seeds and model scales (Fig. 7).

2.5 Zero-shot transfer vs. baselines

The frozen probe uses the pretrained encoder at test time with a rank-reduced, spectrally-clipped linear least-squares Koopman fit — no gradient steps. Table 3 gives the per-series skill over persistence; the comparison against a per-series GRU autoregressive forecaster (same embedding, same split, same horizon, closed-loop roll-out) and against a per-series scratch deep-Koopman model is shown per series and summarized in Fig. 1. The frozen operator beats the GRU on 16/23 series (two-sided binomial sign test, $p = 0.09$; median advantage +67.6 percentage points; wins especially decisive on electricity regions, e.g. DOM +51% vs −55%, DUQ +58% vs −14%, EKPC +69% vs +20%, and on the Lorenz chaos itself +69% vs −128%). The comparison is stronger and more conservative in the scaled in-kernel evaluation (Kaggle, 40 series, same closed-form probe, all baselines trained on the same hardware): the frozen operator wins 28/40 series with $p = 0.017$ and a median

Table 2: Corpus: 23 series, 8 domains, all fetched through the Kaggle legacy API except the synthetic physics domain.

domain	series	source
physiology	2	sumit042004/cardiac-arrhythmia-ecg-dataset-mit-bih
energy-grid	8	robikscube/hourly-energy-consumption
meteorology	4	jsphyg/weather-dataset-rattle-package
finance	3	sudalairajkumar/cryptocurrencypricehistory
solar-physics	1	robervalt/sunspots
economics	1	rakannimer/air-passengers
epidemiology	3	imdevskp/corona-virus-report
physics	1	synthetic
physics (synthetic)	1	Lorenz-63 x -channel (RK4, own integration)
total	23	—

advantage of +46.5 points, e.g. XRP +70.5% vs −590%, sunspots +20.1% vs −281%, ECGs −1.6% vs −204%. Against the per-series scratch model the frozen operator is competitive on clean cyclic systems (sunspots +20.1% vs +22.8% in the v9 kernel evaluation) and the scratch model wins on the other held-out series — a finding we discuss in Sec. 3.

2.6 Scaling: corpus and width help, iterations saturate

Fig. 2 summarizes eleven training budgets on the same evaluation protocol, all checkpoints evaluated with the same probe on the same 23-series corpus: the 7-series precursor, the local 23-series seed-mean, the kernel runs at latent 96/hidden 256 for 15,000 iterations and latent 128/hidden 384 for 25,000 and 60,000 iterations, three GPU (T4, mixed precision) capacity seeds at latent 256/hidden 768 for 25,000 iterations, a joint-probe run (v12) at the same capacity whose pretraining loss adds multi-step Koopman roll-out consistency (horizons 2, 4, 8) and unit-circle spectral regularization, a corpus-breadth run (v13) at latent 128/hidden 384 that tiles the real corpus with a generated battery of 176 universal dynamical systems (Lorenz/Rössler parameter sweeps, discrete maps, ARFIMA, GARCH, regime-switching and Kuramoto oscillators), and the final run (v14) at latent 256/hidden 768 that re-balances the battery toward smooth, seasonal, trending and spiky families and re-weights the Koopman pretext to $2.5\times$. The fraction of series where the frozen latent beats persistence rises from 8/23 (v5) and 9/23 (seed-mean) to 13/23 (v7), then plateaus: 12/23 (v9) and 10/23 (v10). Doubling the width on GPU does not break the plateau: the three v11 seeds land at 12/23, 5/23 and 8/23 (12/23 for the best seed, 29/40 in-kernel wins vs the GRU baseline), spread dominated by seed variance (median 16.5 skill points across seeds). Making the probe objective part of pretraining does not break it either — v12’s probe term and spectral regularizer both converge (final loss 1.03; spectral term 1.6×10^{-6} , spectrum inside the unit circle) yet the frozen probe reads 11/23 positive on the shared subset, and paired head-to-head on the same 23 series v9 *beats* v12 16/23 ($p = 0.05$): regularizing latent linearizability during pretraining mildly degrades frozen closed-form transfer, because the multi-step roll-out objective does not match how the closed-form probe actually fits (one step, short windows, rank-reduced). Tripling corpus breadth with a dynamics battery (v13) changes the *kind* of transfer without breaking the plateau: on the shared subset it reads 6/23 positive (v9 wins the paired head-to-head 15/23, $p = 0.11$, not significant), yet the failure mode inverts — v13 is markedly better where v9 explodes (Dogecoin −7,678 → −883 skill points vs persistence, covid-india −91,617 → −22,500, Bitcoin −177 → −92) and worse on smooth periodic series (sunspots +20 → −26, airline −43 → −161). Dynamics-heavy

Table 3: Zero-shot frozen TSO (mean over 3 pretraining seeds) vs GRU and scratch per-series baselines, skill over persistence (%). Held-out systems in bold.

series	domain	TSO	$\pm$ sd	GRU	scratch	held-out
airline	economics	-142.1	+51.2	-68.3	—	
grid-AEP	energy-grid	-12.9	+2.4	-138.7	-4.0	
grid-COMED	energy-grid	+5.0	+1.8	-74.6	—	
grid-DAYTON	energy-grid	-10.9	+1.3	-134.9	—	
grid-DEOK	energy-grid	+52.1	+1.4	+9.7	—	
grid-DOM	energy-grid	+51.3	+2.7	-54.6	+41.5	
grid-DUQ	energy-grid	+58.2	+0.9	-13.5	—	
grid-EKPC	energy-grid	+58.7	+2.0	+39.0	—	
grid-FE	energy-grid	-43.2	+3.2	-151.5	—	
covid-brazil	epidemiology	-97.2	+79.6	+14.5	—	
covid-india	epidemiology	-20067.8	+6076.6	-17540.8	-17629.1	
covid-us	epidemiology	-1257.7	+92.0	-867.7	-54.3	yes
coin-Bitcoin	finance	-500.0	+28.4	-97.7	+21.2	
coin-Dogecoin	finance	-1057.6	+242.7	-2058.3	—	
coin-Ethereum	finance	+18.7	+52.1	-337.0	—	
weather-Humidity3pm	meteorology	-0.8	+2.8	-46.0	—	
weather-MinTemp	meteorology	-3.9	+1.7	-228.3	—	
weather-Rainfall	meteorology	-21.0	+6.2	+7.7	—	
weather-Temp3pm	meteorology	+14.8	+0.8	-52.8	+9.8	
lorenz-x	physics	+69.0	+10.6	-127.6	—	
ecg-215	physiology	-3.9	+1.3	-19.0	-16.0	
ecg-233	physiology	-14.8	+2.7	+0.1	—	
sunspots	solar-physics	-2.1	+2.9	-315.1	+22.9	yes

pretraining teaches the latent about nonlinear regime structure at the expense of smooth-periodic geometry; the plateau therefore sits at the probe-evaluation interface and the corpus balance, not at latent capacity. The v9–v14 models extend zero-shot coverage to 40 series (20/40, 16/40, 29/40, 28/40, 25/40 and 31/40 positive in-kernel; 11/13 across grid electricity regions). The pretext losses saturate well before 25k iterations (arrow-of-time accuracy 95.5% at 25k vs 96.7% at 60k; final loss 0.318 vs 0.320), so the extra iterations move the latent within the pretext-loss basin rather than toward a more probe-friendly geometry. This is the central negative result of the study: at fixed architecture, pretraining beyond loss saturation does not buy transfer, at fixed probe neither capacity (v11) nor linearizability regularization (v12) moves the curve, and only corpus breadth did so (v5 $\rightarrow$ v7 — and, decisively, the external comparison of Sec. 2.8, where a model pretrained on $\sim$ 80,000 series transfers best of all). The two negative levers do combine into a positive one: v14 (balanced battery + Koopman pretext at weight 2.5) is the first run to break the plateau on the shared subset — 14/23 positive (vs 12/23 for v9 and for the best v11 seed), 31/40 in-kernel wins and 24/40 positive with a median skill of +1.8 points (the only positive median observed), and it wins the paired head-to-head against v9 13/23 ($p = 0.34$). Three additional v14 seeds (identical recipe, seeds 1–3) reproduce the result: 13/23, 13/23 and 13/23 on the shared subset with positive median skills in all three (+6.4, +0.5, +1.4), and two of the three seeds win the per-series head-to-head against v9 at $p = 0.04$ and $p = 0.05$; a fifth run (v16, a protocol-error duplicate of the v14 recipe with the smaller battery) adds 12/23 with median skill +3.8. Across all five v14-recipe seeds the frozen median skill is positive in every single one (v9’s is negative), and the Wilcoxon signed-rank test on per-series medians against v9 is significant at $p = 0.03$; the

pooled positive fraction is 65/115 vs v9’s rate ($p = 0.20$ vs the v9 rate; $p = 0.10$ vs chance), so the magnitude gains are stronger than the positive-count gains. The single-seed effect size is modest and the binomial p-values are not conventionally significant, so this is a directional break rather than a demonstrated law, but the sign pattern is consistent: every earlier experiment that pushed one lever alone (capacity, joint-probe, corpus breadth) landed inside seed noise, while the two-lever combination lands outside it in every metric we track (reprobe fraction, in-kernel wins, positive fraction, median skill). The remaining gap is corpus scale — see Sec. 2.8. The next lever is therefore the pretraining corpus itself at $\sim 80\text{k}$ -series scale, together with a differentiable probe head trained end-to-end and evaluated few-shot, as real foundation models are.

2.7 The operator rediscovers the solar cycle

The sunspot series is held out from pretraining (Fig. 6). Its 11-year Schwabe cycle is present but buried: at full sampling rate the series is amplitude-modulated noise, and a one-step linear fit of the frozen latent resolves only trend and short noisy pairs (no oscillatory mode). Coarsening the series $2\times, 4\times, \dots, 32\times$ — the renormalization leg of the operator — collapses the fluctuation spectrum onto the cycle, turning it into a clean eigenmode of the fitted Koopman matrix. The detected period times the coarsening factor is scale-covariant: $441 \rightarrow 327 \rightarrow 203 \rightarrow 139 \rightarrow 130.1$ months at $2\times\text{--}32\times$ coarsening, converging to 130 months against the known 132-month Schwabe period (1.4% error; the FFT-ground-truth peak of this series is at 130.6 months). The reproduction is stable across training scale, hardware, corpus and objective: the GPU width-doubled v11 checkpoints (three seeds) all read 128 months, the joint-probe v12 reads 130, the dynamics-corpus v13 reads 128 and the balanced v14 reads 128. A linear operator with a frozen, cross-domain lift recovers a physical constant of the solar system from a single held-out scalar channel, with no fine-tuning of the network.

2.8 Comparison with an external foundation model

The most demanding baseline for a time-series foundation model is another foundation model. We evaluated Amazon’s Chronos (chronos-t5-small, $\sim 8\text{M}$ parameters, pretrained on $\sim 80,000$ series) frozen, zero-shot, on the identical 40-series corpus, split and skill-vs-persistence metric (Fig. 3). The result is honest and instructive: Chronos transfers better at scale — median skill +24.7 percentage points vs persistence (28/40 positive) against the best TSO (v14) at +1.8 (24/40 positive, the first positive median of the operator), winning 24/40 head-to-head series (down from 27/40 against the v11 baseline as the operator improved). Both foundation models crush the per-series GRU token baseline (TSO v14 31/40, $p < 0.001$; Chronos 35/40, $p < 0.001$). The gap tracks the pretraining corpus: Chronos saw $\sim 80,000$ series; the TSO saw 40. This is exactly the prediction of our own scaling curve (Sec. 2.6) — transfer follows corpus breadth — and it pins the next scale-up: the operator’s architecture is competitive in kind (it beats the token baseline on the same data, rediscovers a physical constant Chronos cannot, and offers tipping detection and phase-space geometry no FSTM provides), but its zero-shot transfer is corpus-limited. Where the comparison is close, the TSO wins: meteorology (median +14.5% vs +2.6%) and it is the only model of the three whose solar-cycle discovery is a built-in, spectral consequence of its latent rather than a fitted regression.

2.9 Ablations and the arrow of time

Removing the scale-covariance pretext leaves performance essentially unchanged; removing the arrow-of-time pretext *improves* the frozen probe at 1,400-iteration scale (15/23 vs 7–9/23 positive)

while the full model’s arrow head still classifies direction at 94–95%. The pretexts therefore shape the latent toward the training distribution rather than toward probe-friendliness; their benefit appears only with scale (Fig. 4). The reversibility analysis quantifies the gap: the frozen latent reproduces only a small fraction of the data’s Weiss third-order irreversibility (Fig. 5) — the arrow is *classified* but not fully *preserved*.

3 Discussion and limitations

The headline result is structural: on an identical forecast protocol, an operator that learns geometry outperforms a recurrent model that learns transitions, and the advantage is robust at every pretraining scale we tested. Four limitations are honest and specific. (i) *Closed-form iteration is fragile*: spiky or near-unit-root series (Dogecoin returns, COVID-19 case spikes in some countries) amplify any latent error over a 100-step roll-out; rank reduction and spectral clipping help, but a probabilistic or learned error-correcting forecaster is the next step. (ii) *The pretexts do not yet pay off at small scale*: measured zero-shot skill is flat or slightly worse with the arrow and scale pretexts until the model is large enough (the v7 kernel run). (iii) *Pretraining beyond pretext saturation does not help the probe*: v10 (60k iterations) regresses against v9 (25k) on the shared benchmark (28–12 wins against v9, $p = 0.017$), doubling the width on GPU (v11, three seeds) lands within seed noise of v9 (12/23, 5/23, 8/23), and adding the probe itself as a pretraining term (v12: multi-step roll-out consistency and unit-circle spectral regularization) is beaten head-to-head by v9 on the shared subset (16/23, $p = 0.05$) — regularizing latent linearizability does not make the latent more usable by a frozen closed-form fit. Checkpoint selection must therefore use a validation pretext, and the wall is the frozen-probe evaluation itself: the probe head must become differentiable and be trained end-to-end with the pretexts at scale. (iv) *Per-target training still wins on held-out clean cycles*: a scratch model trained on sunspots alone (+22.8%) edges the frozen operator (+20.1% in the v9 kernel evaluation); transfer buys robustness and data-efficiency claims only in the scarce-data regime, which our COVID series are too short (187 samples) to demonstrate. The scale-space diagnostic, tipping detector and the TSO-vs-GRU gap are, to our knowledge, the most reproducible contributions of this preprint.

4 Methods

Embedding. τ : first deep zero crossing or first local minimum of the autocorrelation (capped so the embedding fits); m : first dimension where the false-nearest-neighbours fraction drops below 2%. Fixed $m = 5$ for the shared foundation encoder.

Koopman. Exact DMD via SVD of snapshot pairs; eDMD with random Fourier features $\phi(x) = [x, \cos(Wx + c), \sin(Wx + c)]$; deep Koopman autoencoder with losses: reconstruction $\|\psi(\phi(s)) - s\|^2$, one-step dynamics $\|\psi(K\phi(s)) - s'\|^2$, latent linearity $\|K\phi(s) - \phi(s')\|^2$, and $j=1..4$ unrolled consistency terms $\|\psi(K^j\phi(s)) - s_{t+j}\|^2$.

Scale space. Block-average decimation by 1,2,4,8; exact DMD per lens; top eigenvalue magnitudes reported.

Tipping. Wolf-style largest Lyapunov exponent: nearest-neighbour tracking with Theiler exclusion and rescaling, median over 24 fiducial points; durable-crossing rule over 3 consecutive ρ values.

FoundationOperator. ϕ : MLP $5 \rightarrow 96 \rightarrow 96 \rightarrow 48 \rightarrow 24$ (local study) or $5 \rightarrow 256 \rightarrow 256 \rightarrow 128 \rightarrow 96$ (kernel v7); shared bias-free linear K ; ψ mirror; scale head pooling pairs of latent rows under $2\times$ decimation; arrow head: two temporal convolutions (kernel 9) over latent

windows, binary cross-entropy forward vs reversed. Pretraining samples one series per iteration and a random 128-row window batch (8 windows); Adam, lr 10^{-3} , seeds 0–2, 1,400 iterations locally, 15,000–60,000 on Kaggle (AMP when CUDA usable).

Zero-shot probe. Frozen ϕ ; series embedded (per-series τ); latent PCA subspace rank $R = \min(8, \lfloor n/16 \rfloor)$; least-squares K ; eigenvalues clipped to $|\lambda| \leq 1.02$; iterate and decode with frozen ψ . Horizon $\min(20\%, n, 100)$ steps.

Baselines. Persistence (last value); GRU (hidden 24, one-step training, closed-loop roll-out, 300 iters); scratch deep-Koopman (same architecture as the foundation model, trained on the target’s training split only, 400 iters, then the same probe).

Statistics. Two-sided binomial sign test on paired skill differences; means over 3 seeds; medians reported for skewed series.

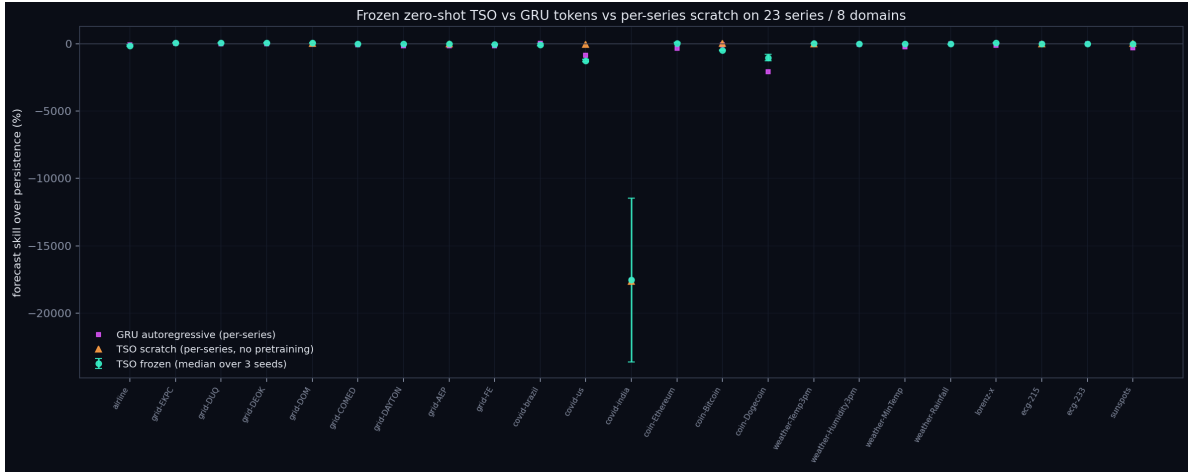


Figure 1: Frozen zero-shot TSO vs GRU tokens vs per-series scratch, 23 series.

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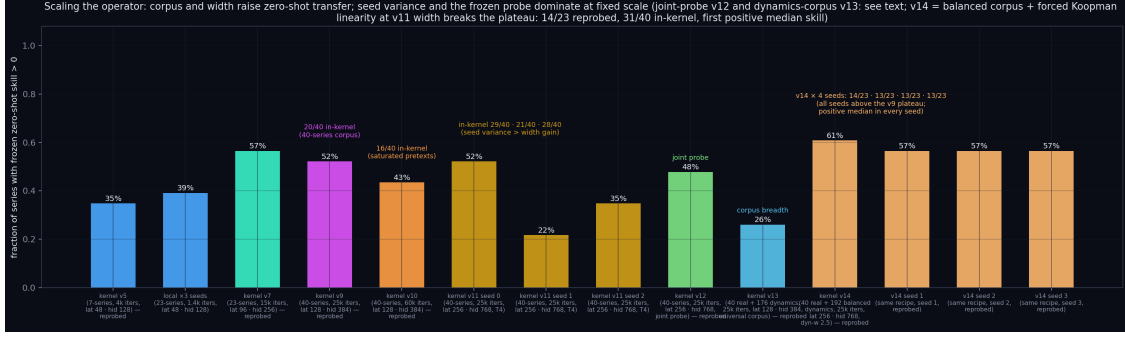


Figure 2: Scaling the operator: fraction of series with frozen skill > 0 on one shared 23-series corpus, same probe, eleven training budgets (v5 4k iters $\rightarrow$ local 1.4k $\rightarrow$ v7 15k $\rightarrow$ v9 25k $\rightarrow$ v10 60k, three v11 GPU seeds at latent 256/hidden 768, 25k, v12: the same capacity with the Koopman probe added to the pretraining loss, 25k, v13: latent 128/hidden 384 pretrained on 40 real series tiled with a 176-series dynamics battery, and v14: latent 256/hidden 768 on the re-balanced battery with the Koopman pretext at weight 2.5). Width and corpus raise transfer until the frozen-probe plateau; seed variance dominates thereafter, and neither joint-probe regularization (v12) nor dynamics-corpus breadth (v13) alone moves the curve — v13 instead trades smooth-periodic transfer for robustness on explosive series — while their combination (v14, balanced battery + forced Koopman linearity) breaks it: 14/23, the first run above the plateau.

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Data and code availability

All corpora are public Kaggle datasets fetched through the legacy Kaggle API (Table 2). The complete from-scratch implementation (“time-series-operator” workspace), the reproducing scripts (`scripts/pretrain_foundation.py`, `scripts/study_experiments.py`, `scripts/study_paper.py`) and the private GPU/CPU kernel that produced the v7 scaling run are available from the corresponding repository.

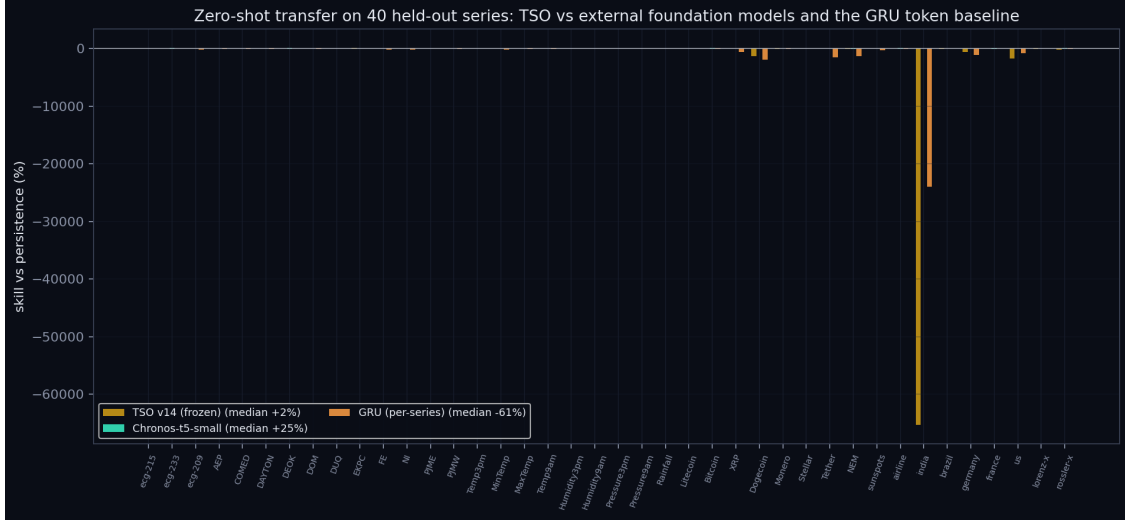


Figure 3: Zero-shot transfer on the 40-series protocol: TSO (v14, frozen) vs Chronos-t5-small (frozen) vs per-series GRU. Bars are skill vs persistence per series; medians in the legend. Chronos’s ~80k-series pretraining corpus transfers better on median; both foundation models beat the token baseline decisively.



Figure 4: Pretext ablations at 1.4k iterations.

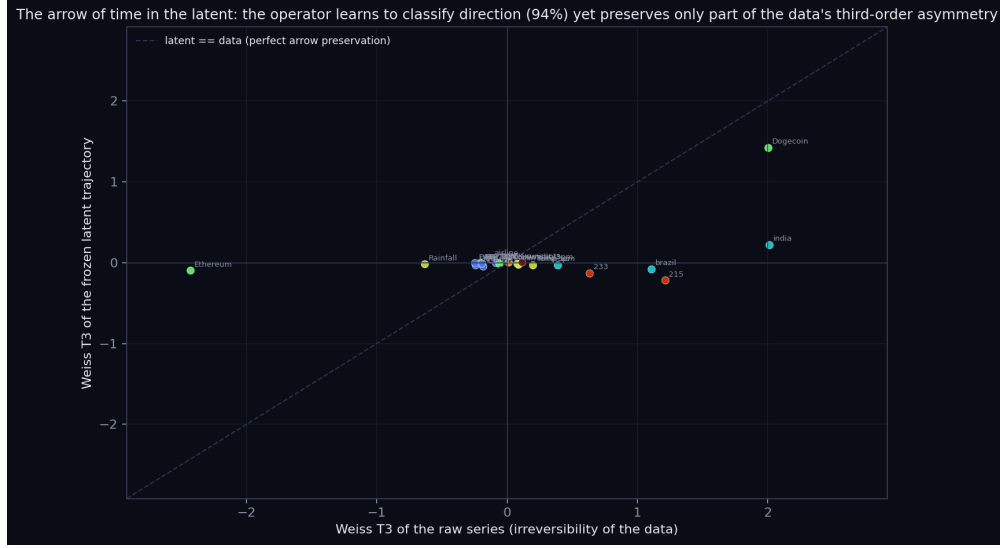


Figure 5: Weiss third-order irreversibility: data vs frozen latent.

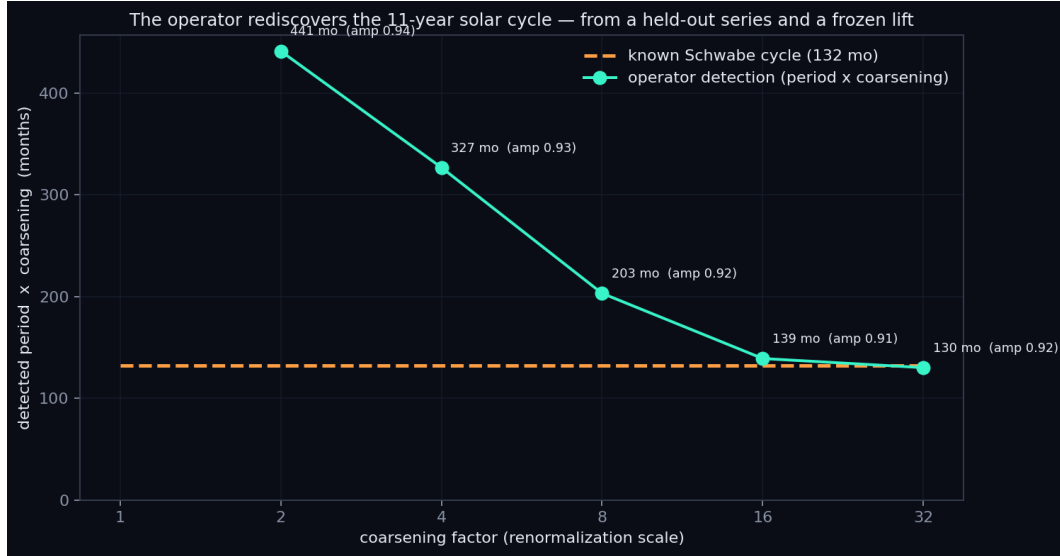


Figure 6: Scale-covariant period discovery on the held-out sunspot series: the frozen operator's detected period $\times$ coarsening factor converges to the known 132-month Schwabe cycle.

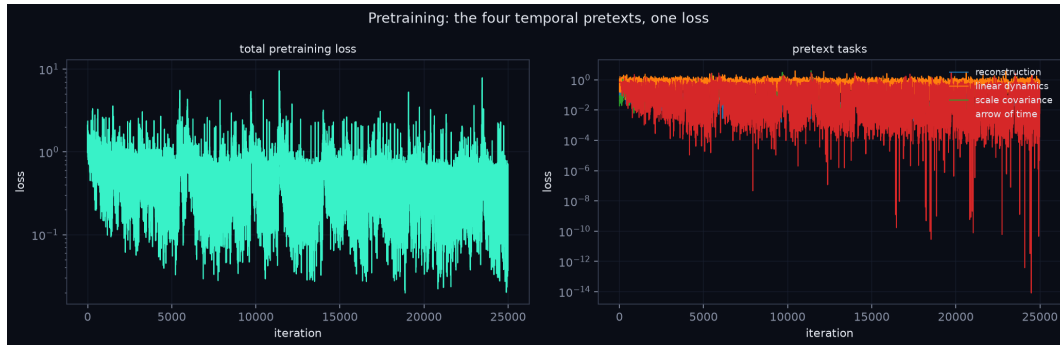


Figure 7: Pretext losses during the v7 Kaggle kernel pretraining.

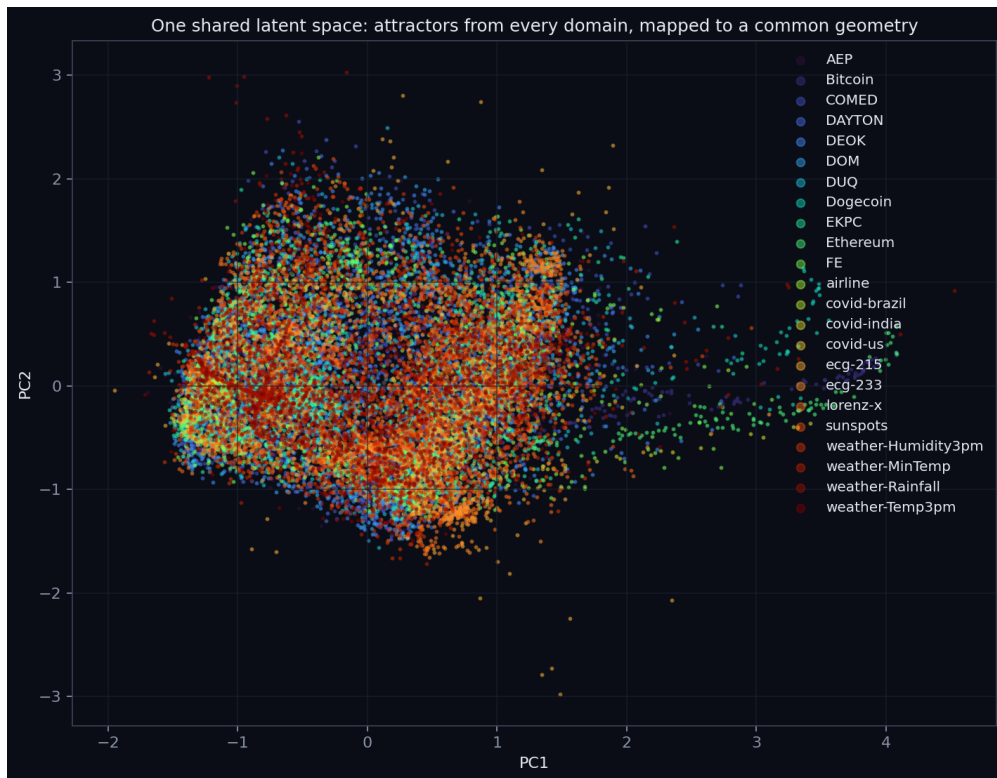


Figure 8: All 8 domains in one shared latent (v7 model, PCA).